symbiOS: Minimal Sovereign Informational Kernel

Abstract

This paper presents symbiOS, a sovereign informational kernel designed for high coherence and security. It introduces the concepts of Mem-Nano-Bit (MNB) as a minimal unit of auditable semantic memory and COER (Cor-Nano-Bit) as a coherence engine governed by mathematical invariants (Ψ, Ω, CVaR, PoLE).

Core Architecture

The system is built on three sovereign layers:

1. **Scientific Layer**: Formal metrics and thermodynamic mirror concepts.
2. **Normative Layer**: Laws and invariants governing state transitions.
3. **Runtime Layer**: Executable environment for the kernel.

Security^2 (S^2)

Security^2 protects the security logic itself, metrics, and governance against internal corruption and semantic capture.

- **Metric Security**: Deterministic and auditable calculation of Ψ, Ω, CVaR.
- **Invariant Security**: Ω-GATE verification before any state write.
- **Thermodynamic Security**: Asymmetric cost for attackers vs. system adaptation.

Implementation

The kernel uses a 3-body persistent memory model. The state is represented geometrically as an organic cube where:

- **Purple/Violet**: High coherence (High Ψ).
- **Blue**: Normal operation.
- **Red**: Critical state.

Conclusion

symbiOS transforms coherence into infrastructure, providing a measurable, auditable, and publishable framework for autonomous organisms.